



Effects of different tree species on infiltration and preferential flow in soils developing at a clayey spoil heap

Lukáš Jačka^{a,*}, Alena Walmsley^b, Martin Kovář^a, Jan Frouz^c

^a Department of Water Resources and Environmental Modeling, Faculty of Environmental Sciences, Czech University of Life Sciences Prague, Kamýcká 129, Praha –Suchbát 165 00, Czech Republic

^b Department of Landscape and Urban Planning, Faculty of Environmental Sciences, Czech University of Life Sciences Prague, Kamýcká 129, Praha–Suchbát 165 00, Czech Republic

^c Institute for Environmental Studies, Faculty of Science, Charles University, Benátská 2, Prague 2, 128 01, Czech Republic

ARTICLE INFO

Handling Editor: Yvan Capowiez

Keywords:

Forest reclamation
Redistribution of infiltrated water
Macropore flow
Brilliant blue
Hydrological effects of trees

ABSTRACT

Soil recovery is crucial for successful ecosystem restoration at post-mining sites. Water infiltration and preferential flow in soil are important but little studied soil characteristics. In this study, we explore effects of tree species on infiltration, geometry of preferential flows, size distribution and origin of pores, and related properties of soil. An experiment was conducted at post-mining sites developing under monocultures of three tree species (alder, oak and spruce) after lignite mining near Sokolov, Czech Republic. A key hydraulic parameter of soil, field saturated hydraulic conductivity (K_s), measured in topsoil, was highest under alder (mean value of $6.9 \times 10^{-5} \text{ ms}^{-1}$) and lowest under spruce (mean value of $1.9 \times 10^{-5} \text{ ms}^{-1}$). In subsoil, there were no significant differences in K_s between tree species. For the alder site, dye tracer tests showed that water penetrates almost uniformly to the top 15 cm of soil depth and with the greatest lateral flow. In a deeper part of the alder soil profile, preferential flow was concentrated around earthworm burrows and roots, where the two often occur together. Thin soil sections of alder soil show that earthworm casts and burrows substantially contribute to porosity around the roots (59 % of dyed worm-created structures in alder-root microsite). A similar situation was observed in oak, but preferential flows along roots started from the surface and cross section of preferential flow was smaller than in alder soil. In spruce, preferential flow was concentrated around roots and cracks in claystones and the cross section was even smaller. Comparison among the forest stands indicates that overall porosity does not correspond with infiltration of dye tracer and K_s , which seem to be more driven by pore size distribution and pore connectivity. This is consistent with larger occurrence of macropores ($>300 \mu\text{m}$) in alder. The results show that trees affect infiltration and preferential flow in soil by their specific root systems and to a large extent indirectly by effect of litter quality on earthworm populations, which then enhance macroporosity and pore connectivity.

1. Introduction

Open-cast mining causes extensive disturbances, destroying entire ecosystems that are either excavated or buried under a deep layer of overburden material. This type of mining changes not only individual ecosystems but whole landscapes (Bradshaw, 1983; Bradshaw, 1997; Macdonald et al., 2015). Therefore, rehabilitation of such an area is crucial for restoration of both the ecosystem functions and production functions of the post-mining landscape (Bradshaw, 1983; Frouz, 2014; Macdonald et al., 2015).

The water regime of the entire area is wholly disrupted by the mining activities, with waterways being diverted and topography of the

landscape being altered (Bonta et al., 1997). The piling up of the external spoil heaps leads to compaction of the geological profile and consequently water from the belowground aquifer is pushed to the surface, where it creates lakes and swamps (Roják, 2013). On top of the spoil heaps, however, only little water is stored, due to a lack of soil and overburden compaction. Previous studies have shown that the freshly heaped material differs overwhelmingly from developed soil, often having extreme pH and extreme texture consisting of a narrow range of soil particle sizes, being either too sandy or clayey. Due to its lack of structure, this material can exhibit unfavorable hydrophysical properties and infiltration characteristics and consequently be subject to erosional processes if left unreclaimed (Kuráž, 2001; Váška et al., 1993).

* Corresponding author.

E-mail address: jacka@fzp.czu.cz (L. Jačka).

<https://doi.org/10.1016/j.geoderma.2021.115372>

Received 13 December 2020; Received in revised form 23 July 2021; Accepted 25 July 2021

Available online 7 August 2021

0016-7061/© 2021 Elsevier B.V. All rights reserved.

Especially in materials with low permeability (i.e., if overburden contains high percentage of the fine particles - silt and clay), infiltration can be insufficient. Consequently, low infiltration capacity limits potential water storage in soil because infiltration determines how much water enters the soil and is then distributed within the soil profile (Kutfllek and Nielsen, 1994; Hillel, 1998). Infiltration divides rainfall into surface runoff and the part that enters the soil profile (Horton, 1933). Therefore, it is a key hydrological process that enriches the soil profile with water (Philip, 1957). Saturated hydraulic conductivity, K_s , is a crucial dynamic hydraulic parameter describing water flow in soil (Reynolds et al., 2000; Jačka et al., 2014), and therefore it is an essential parameter describing the infiltration process (Jačka et al., 2016).

As soil water flows in soil pores, a whole complex of quantitative and qualitative characteristics of the porous system (e.g., content, size, tortuosity, connectivity, and coating of pores) drives infiltration and distribution of infiltrated water through the soil profile (Gerke and Köhne, 2002; Gerke, 2006; Soracco et al., 2019; Zhang et al., 2020). Key roles are played by macropores and upper size capillary pores, where the dominant part of infiltrated water can be distributed by preferential flow, if these pores are open and interconnected (Beven and Germann, 1982; Alaoui and Helbling, 2006; Ali et al., 2018). Soil macropores can be divided into 1) biopores (created by soil fauna activity and roots), and 2) pores related to soil aggregation, such as cracks in clay soil created by swelling and shrinking phenomena (Jarvis, 2007). All these macropores can be interconnected and create a complex system in soil. Apart from the macropore flow, other subsets of preferential flow can occur in soil, such as fingering, unstable flow, funnel flow, and heterogeneous flow (Gerke, 2006; Jarvis, 2007).

By means of preferential flow, water can be quickly distributed through the entire soil profile, even to the deeper layers (Alaoui et al., 2011; Tang et al., 2020). Geometry of preferential flow strongly affects which parts of a soil profile are in contact with infiltrated water. Critically important for water accumulation and retention in a soil profile is water transfer among macropores and soil matrix, which is controlled by connectivity and interface properties of these two domains (Beven and Clarke, 1986; Weiler and Naef, 2003; (Weiler and Flüeler, 2004)). Brilliant Blue FCF dye tracer can be advantageously used for visualizing the preferential flow of infiltrated water and also to show connectivity of a porous system (Schwen et al., 2014; Jačka et al., 2018; Zhu et al., 2019), although this method also exhibits some deficiencies, e.g. problematic sorption in some cases and lower mobility than untreated water in smaller-sized pores (Nobles et al., 2010; Wang and Zhang, 2011).

Despite there being some research available about water retention development in post-mining sites and some understanding of how water retention corresponds with properties of trees and soil biota community (Frouz and Kuráz, 2014; Cejpek et al. 2013; Cejpek et al., 2018), there is very little data about development of water infiltration and preferential flow paths at post-mining sites. There exists a large body of evidence that vegetation cover (and especially trees, due to their size and lifespan) can have major impacts on soil water regime. This impact is not only through transpiration and interception, but also through the root architecture, litter produced and its transformation in the soil, as well as via indirect effects on soil fauna (Frouz et al., 2013; Bargañés Tobella et al., 2014; Benegas et al., 2014; Walmsley et al., 2019). Channels around the root systems of plants, which are specific for each species, distinctly affect macroporosity, preferential flow, and infiltration (Devitt and Smith, 2002; Jačka et al., 2012; Luo et al., 2019; Zhu et al., 2019). Litter of coniferous and some deciduous trees can be water-repellent, and, as a consequence, the preferential flow and amount of water infiltrated into soil is influenced (Bens et al., 2007). The quality of tree litter affects water regime of the soil also indirectly, as it determines the characteristics of the soil humus layer (Tomlin et al., 1995; Frouz et al., 2006). Deciduous and in particular broadleaf trees producing litter with low C/N ratio sequester soil organic matter faster than do trees producing litter with high C/N ratio (Frouz et al., 2009, 2013; Frouz, 2018). It also has been documented that carbon sequestration at post-mining

sites revegetated with deciduous trees is largely supported by soil fauna (especially earthworm) activity (Frouz et al., 2009; Frouz et al., 2013).

Earthworms not only speed up humus formation at rehabilitated sites, they also increase the quantity of macropores while changing pore size distribution and soil structure by drilling, creating biopores, and bioturbation (Scullion and Malik 2000; Marashi and Scullion 2004; Frouz et al., 2006; Frouz et al., 2007; Frouz et al., 2013; Cheng et al., 2017; Frouz, 2018). Therefore, the worms enhance soil's preferential flow and infiltration capacity (Capowiez et al., 2014). This effect differs depending on the earthworm ecological group, as these groups vary in burrow characteristics and feeding behaviors. Epigeic earthworms, which feed purely on the litter and generally don't venture beyond the organic layer of soil, contribute to litter decomposition. Consequently, these earthworms increase the active soil surface, which increases water storage capacity of the topsoil (Ernst et al., 2009). The vertical burrows of anecic species, on the other hand, increase water infiltration into soil and water transport into deeper soil layers (Capowiez et al. 2015; Hallam and Hodson 2020). Endogeic earthworms create semi-permanent horizontal burrows in the upper mineral layer of soil, which are often partially filled with their excrements. Their burrowing and feeding activity increase the water holding capacity of the humus layer of soil and affects horizontal water distribution in this layer as well (van Schaik et al., 2014; Hallam and Hodson 2020).

Less explored is the relationship between earthworms and roots and their joint effect on soil porosity. Kautz et al. (2014) have found for perennial crops that earthworms often utilize macropores created by roots (old root channels). Because tree roots exert greater pressures than soft earthworm bodies (Ruiz et al., 2015), the roots are expected to penetrate the hard clay slabs initially, thereby enabling anecic earthworms to reach the deeper soil layers.

Covering post-mining sites with topsoil can substantially improve soil properties (Borůvka et al., 2012). However, there generally is not enough material available to cover the whole area and trees are often planted into bare overburden. This is also the case of post-mining forest sites in the Sokolov District, in northwest Czechia, where the present study was done. Overburden at these sites consists of tertiary clay material, which is rich in mineral nutrients and slightly alkaline (Frouz et al., 2013). Soil recovery after the reclamation of this material can be rapid and strongly influenced by tree species planted (Frouz, 2018). The examined spoil heap therefore presents unique opportunity to study the sole effect of individual tree species on soil development. Due to the soil's adverse hydrophysical properties and disturbed water regime in the spoil heaps (Kuráz, 2001; Frouz and Kuráz, 2014), the effects of forest reclamation on crucial soil hydrological characteristics, namely infiltration and preferential flow, should be studied deeply. These effects can distinctly influence recovery of the ecosystem as a whole.

The main aims of the current study are to explore differences in 1) field saturated hydraulic conductivity (K_s) measured by Guelph permeameter (GP), 2) related physical properties (content and size of pores, bulk density) using undisturbed soil samples and 3) preferential flow in the soil profile using dye tracer infiltration tests for three specific soils. These soils were developed at post-mining sites under three tree species (monoculture of alder, oak, and spruce) having different litter quality as characterized by C/N ratio (Frouz et al., 2013) and different root systems. The major hypothesis is that soil under tree species characterized by more easily decomposable litter and lower C/N ratio (and as a consequence higher abundance of earthworms) will exhibit higher K_s -value, infiltration and preferential flow. Based on the findings of Walmsley et al. (2019), who sampled earthworms in the area in the previous year, we assume that the highest counts of earthworms and diversity of species will be in alder stand, followed by oak and the lowest numbers will be in spruce. However, to confirm this, we have sampled earthworms at the three stands used in our study. The effects of biostructures (worm-created structures and root-created structures) and cracks in clay on distribution patterns of the dyed water throughout the

soil profile were visually and microscopically analyzed using the intercept method (modified from Frouz et al., 2007; see Supplement for details). To evaluate dye tracer distribution in macroscale, the intercept method was applied on profile photographs. To evaluate dye tracer pathways and their connectivity in detail, the intercept method was applied in microscale on thin soil sections containing biostructures. Therefore, to obtain more comprehensive hydrological view, effects of trees were investigated on three different scales: 1) standard point scale (GP tests and undisturbed samples), 2) pedon scale (dye tracer distributions on profile photographs), and 3) microscale (dye tracer distributions on thin soil sections).

2. Material and methods

2.1. Study area

The studied Vintřovská spoil heap (a part of the larger Velká Podkrušňohorská spoil heap) is located in northwest Czechia's Sokolov mining district. Average altitude of the study area is 500–600 m a.s.l., mean annual precipitation is 650 mm, and mean annual temperature is 6.8 °C. The examined spoil heap consists of slightly alkaline tertiary Miocene claystones (clays of the cypress series), which form overburden during lignite mining. The major clay minerals are kaolinite, illite, and montmorillonite accompanied by quartz and carbonates, impregnated by fossil organic matter (Kříbek et al., 1998; Frouz et al., 2001). The substrate is well supplied with mineral nutrients and basic cations; clay content in the deposited claystone is about 70%; and pH of the substrate in initial successional stages is in the range 8–9 (Kuráz et al., 2012). Mining out individual layers of sediments and their mixing during deposition created rather homogeneous patches of deposited material with areas of several hectares (Frouz et al., 2011). One such homogeneous patch was used in our study.

When deposited, overburden had the character of solid pieces of mudrock, specifically claystones, of various sizes with the biggest being

about 60 × 10 cm. Thereafter, during the next several winters, the claystones cracked into layers that reflect the sedimentation process in geological times. Although these structures still look to be solid pieces of stone, they can easily break into lamellar structure when force is applied. If mudrocks are then levelled by earthmoving machinery, they are broken to smaller pieces. The smaller pieces and edges of the stones also turn into amorphous clay during wet periods. The clay thus released fills spaces between the solid pieces and forms a soil layer with low water permeability (Frouz et al., 2018).

This study benefits from the existence of what may be termed a “common garden experiment” created by planting several tree species in small, monoculture patches (ca 0.2–2 ha each), resembling a chessboard or patchwork pattern. This allows us to study soil properties developed under various tree species at sites located close to one another on the very similar substrate. Other details of this specific study area have been described by Frouz et al. (2013) and Walmsley et al. (2019).

2.2. Study sites and sampling design

Three study sites were selected on the Vintřovská spoil heap (see Fig. 1). Each experimental site was a rectangle (ca 10 × 20 m) and was dominated by a single tree species: alder (*Alnus glutinosa*), oak (*Quercus robur*), or spruce (*Picea abies*). These tree species were selected due to their production of litter with different qualities. According to Frouz et al. (2013), the mean values of C/N ratios for the fallen litter at the Vintřovská spoil heap were 24, 35, and 52, respectively, for the alder, oak, and spruce. These ratios for green leaves were 18, 46, and 79, respectively, for the alder, oak, and spruce. The sites were adjacent (see Fig. 1) and were all characterized by the same substrate and age (planted in 1978). Therefore, selection of tree species for forest reclamation was probably the only factor causing soil development and soil characteristics in the selected study sites.

To characterize bulk density and pore size fractions of the topsoil, undisturbed samples were collected immediately below the litter layer.

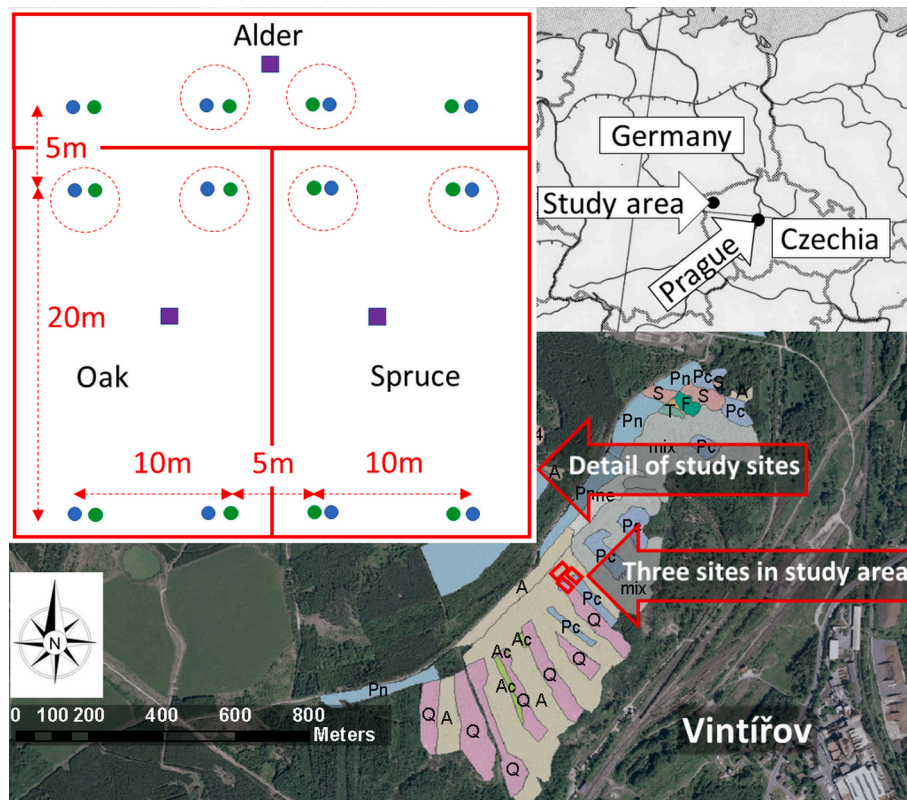


Fig. 1. Location of the study area (Vintřovská spoil heap) within Central Europe, positions of the three study sites (red rectangles) on the reclamation patches (modified from Walmsley et al. (2019)), and detail of the sampling positions on the sites. Abbreviations denote tree species of the patches (A - *Alnus* spp., Ac - *Acer pseudoplatanus*, F - *Fraxinus excelsior*, mix - mixed stands, Pn - *Pinus* spp., Pc - *Picea* spp., Q - *Quercus robur*, S - *Salix caprea*, and T - *Tilia cordata*). The points denote positions of Guelph permeameter tests performed in the topsoil (green) and subsoil (blue), red dash-line circles denote positions where three undisturbed samples were collected, and the purple squares denote dye tracer tests.

Field saturated hydraulic conductivity (K_s) was measured using Guelph permeameter (GP) in shallow wells flooded to the level, which allowed to sample topsoil immediately below the litter layer. Therefore, for both methods, the topsoil was sampled at the lowest practicable sampling depth below the soil surface. To characterize subsoil, undisturbed samples were collected at mean sampling depth of ca 28 cm (vertical depth to the ring center from the soil surface) and GP tests were performed in wells of ca 28 cm depth. These depths for subsoil were selected in order to always sample below the drilosphere/humus or fermentation/humus layer (see thicknesses of these layers in Fig. S1).

Overall number of GP tests was 24 (3 sites $\times$ 4 replicates $\times$ 2 soil profile positions - topsoil and subsoil, see Fig. 1). Sampling positions of GP tests were selected in such a way that for all replicated measurements the closest replicate located in a different tree species was much closer than the most distant replicate in the same tree species (see Fig. 1). These sampling positions were selected to minimize a risk of potential spatial pseudo-replications. Minimum distance between wells was established at 1.5 m to avoid interaction of GP tests. Two to three undisturbed samples were collected close to each GP test (overall, $3 + 3 + 2 + 2 = 10$ samples in topsoil and the same number in subsoil at each site) to measure bulk density and to estimate pore size distribution. Positions, where the three undisturbed samples were collected close to the GP tests, are depicted in Fig. 1 by red dash-line circles. Overall number of undisturbed samples was 60 (3 sites $\times$ 10 replicates $\times$ 2 soil profile positions - topsoil and subsoil). One single-ring infiltration test with dyed water was performed in the center of each study site.

Field work was restricted to two consecutive days (4–5 June 2018) in order to eliminate possible temporal variation of the examined soil properties. Air humidity was 71 and 73 %, temperature was 19.5 and 18.9, respectively, on the 4th and 5th June (average daily values). There was no precipitation on these days. Daily precipitations and temperatures (1 May–5 June 2018) are shown in Fig. S6. Initial volumetric soil water contents (percentages, mean $\pm$ SD) determined for topsoil were the following: alder – 35.2 ± 5.0 , oak – 43.1 ± 3.2 , spruce – 30.1 ± 5.2 . For subsoil, these values were the following: alder – 40.8 ± 4.0 , oak – 44.6 ± 3.44 , spruce – 35.3 ± 5.1 .

2.3. Field measurements of K_s

The Model 2800 K1 GP kit (Soilmoisture Equipment, Goleta, CA, USA) was used to measure K_s . The standard single constant head method for GP was applied according to Soilmoisture's operating instructions (Soilmoisture, 2012). Each GP test was made in an individual well ca 6 cm in diameter which was flooded to a constant water level of 5 cm. Sorptive number 0.12 cm^{-1} was estimated on the basis of visual estimate of the clay soil structure according to Elrick et al. (1989). Well-shape factor was calculated according to Zhang et al. (1998). For K_s calculation, the equation given by Reynolds (2008) was used. This equation is also described by, for example, Jačka et al. (2018).

2.4. Laboratory measurement of bulk density, soil water content, pore size fractions

Measurements were performed on undisturbed soil samples collected into standard Eijkelkamp stainless steel rings (volume, 100 cm^3 ; inner diameter, 5.6 cm; height, 4.06 cm; Eijkelkamp, Netherlands). The following procedure was used for estimating volumetric fractions of pore sizes in selected ranges. Points of soil water retention curves (SWRC) of undisturbed samples were measured using Sandbox (Model 0801, Eijkelkamp, Netherlands) according to the standard method as described by Eijkelkamp (2019). Suction pressures (i.e., negative pressure heads) of 0, 10, 32, and 100 cm were applied in Sandbox. Volumetric fraction of drained pores (=decrease in volumetric water content) was determined for two consecutively applied levels of suction (i.e., 0 and 10, 10 and 32, and 32 and 100 cm). Zero value of suction corresponds with full possible saturation of soil with water. Total

porosity estimate was considered to be equal to saturated water content.

Values of the applied suction pressures were used to estimate ranges of pore sizes (equivalent pore diameters) according to the following procedure. For SWRC measurement, applied suction pressure is practically equal to the capillary forces, which hold water in the soil sample. From the value capillary forces (expressed as height of the capillary rise of water), the equivalent pore diameter was calculated using the well-known Young–Laplace equation. For description of the calculation see e.g. Lim et al. (2016), Hillel (1998), or Kutilek and Nielsen (1994).

After SWRC measurement, undisturbed soil samples were oven-dried to a constant weight. Estimated total porosity was considered to be equal to the saturated water content, which was calculated from the difference in weights between saturated soil (0 cm suction) and oven-dried soil. Bulk density was calculated using mass of dried soil (in grams) divided by total volume of the sample (100 cm^3). Soil water content (to estimate initial condition before infiltration test) was determined gravimetrically from the difference in weights (collected undisturbed sample in the field minus oven-dried one).

2.5. Dye tracer infiltration tests

One single-ring infiltration test was done at each study site in order to visualize the preferential flow of dyed water in the soil profile. A Brilliant Blue FCF dye tracer was applied in a water solution of 5 g l^{-1} (Schwen et al., 2014; Jačka et al., 2018). A single-ring infiltrometer (30 cm inner diameter) was inserted into the soil to the depth of 4 cm (position of the lower sharpened edge of the ring below the surface). To avoid flow in soil-surface interface due to irregularities of the terrain, this depth was selected as the lowest possible. All fissures in soil around the wall of the ring created during the insertion were thoroughly sealed with clay. A total of 24 L of dyed water were applied during each test. The dyed water level of ca 4 cm (practically constant pressure head) was maintained inside the infiltrometer by gradual refilling and a reference spike (see Jačka et al. (2016) for a detailed description).

2.6. Analysis of dyed patterns - macroscale and microscale

Vertical and horizontal cross sections were excavated in the soil profile ca 24 h after infiltration of all dyed water. Vertical cross sections were excavated at various horizontal distances (progressively from 80 to 0 cm) from the center of the ring. Therefore, the final vertical cross section was made through the center of the infiltration area. Further, horizontal cross sections were excavated on the second half of the infiltration area at various vertical depths below the soil surface (progressively from 3 to 25 cm). Excavated cross sections were visually inspected and dyed patterns in soil were measured with a tape measure and described. Additionally, photos of these cross sections were taken to document size and intensity of preferential flow. Photos were taken perpendicular to each cross section in order to cover the whole excavated area. A scale bar was present in each photo.

Photos of the vertical cross sections were used to evaluate presence of dye in individual macroscopically distinguishable features. To do so, an intercept method was used. We projected a regular grid of points on each photo. At each point we checked if the point hit a stained structure or if there was no dye there. If the point hit stained structure, we determined whether this structure was a visible biostructure (i.e., root-created structures – rhizosphere or worm-created structures – drilosphere) or a macropore in the soil matrix (crack in clay). More details about intercept method are given in Supplement and Fig. S2.

Drilosphere can be defined in narrower or broader sense (Andriuzzi et al., 2013; Brown et al., 2000; Bouché, 1975). Further, for purposes of our research, we consider drilosphere as worm-created structures observed within the examined soil profiles, i.e. casts, burrows (type of macropores) and linings around burrows. According to Helliwell et al. (2017), rhizosphere is the zone of soil around the root which is affected by plant activity. Further, for hydrological purposes of our research, we

consider rhizosphere as pores in soil created along the roots (channels around roots).

After taking photos of individual vertical cross sections, we took three soil monoliths of $5 \times 5 \times 10$ cm per each microsite type that was identified on the cross section. Soil monoliths were taken in a way that included three different microsite types – non-stained clay overburden (hereafter called clay-microsite) and dye-stained macropores formed either by visible root or earthworm burrow (hereafter called root-microsite and burrow-microsite, respectively). This means that in total we had nine monoliths for each dye tracer infiltration test, three for each microsite type, with the exception for spruce stand, where no stained earthworm burrows were found and sampled. Monoliths were air-dried, impregnated by epoxy resin and then cut using diamond tools into 5×5 cm thin soil sections, 1 cm apart. The thin section from the middle of the monolith covering the selected microsite was used for future analysis using the intercept method. Each thin section was observed under a binocular microscope. For each observed stained structure we also made notes regarding the extent to which it was connected with other structures and how they were oriented. See Supplement and Fig. S2 for more details about the intercept method and Frouz et al. (2007) for more details about the thin soil section preparation.

2.7. Earthworm sampling

At each study site, four soil cores (30×30 cm square area, 25 cm deep) were excavated to determine the abundance of earthworms. The cores were 5 m from one another in a square grid, located approximately at the center of the study site. All the earthworms were extracted by hand-sorting, counted, preserved and later determined down to species or order. The method of earthworm sampling is further described by Walmsley et al. (2019), who had studied earthworms at these sites in the previous year. Sampling of earthworms served to verify findings of Walmsley et al. (2019) and Frouz et al. (2013) who observed distinct differences in abundance of earthworms among alder, oak and spruce stands at Vintřovská spoil heap.

2.8. Statistical analysis

Statistical analyses were performed in the R software environment (statistical test in Stats package, version 3.0.2; and descriptive statistics in Pastecs package, version 1.3–15). All statistical tests were performed at significance level = 0.05. Data for the selected soil property (K_s , bulk density, contents of equivalent pore size range) measured at one of the three sites (alder, oak, or spruce stand) and at the same position (topsoil or subsoil) were treated as an individual data set. Descriptive statistics and normality tests (Shapiro and Wilk, 1965) were calculated for each of these data sets. Next, for topsoil and subsoil separately, one-way analysis of variance (ANOVA) was used to assess differences in the selected soil properties (K_s , bulk density, contents of equivalent pore size range) among the three forest stands. If ANOVA indicated significant difference, Tukey's honestly significant difference test (Tukey's HSD test) was applied for multiple pairwise comparisons of means. In addition, Welch's two-sample *t*-test (*t*-test) was used to assess differences between means of the selected soil property among topsoil and subsoil at each forest stand. For earthworm abundance and intercept method analysis, due to lower numbers of replicates (four for earthworm sampling, three for intercept method), we calculated only means and standard deviations and we have not tested differences in means.

3. Results

3.1. Saturated hydraulic conductivity at the three forest stands

Significant differences in K_s were noted in topsoil (*p*-value of the ANOVA test = 0.006). For alder stand in topsoil, the mean value of K_s was significantly higher than for oak and especially so for spruce stand,

where the lowest value was measured (see Fig. 2). For subsoil, no significant differences among tree monocultures were observed (*p*-value of the ANOVA test = 0.334). For each tree stand, K_s values measured in topsoil were higher than those in subsoil (see Fig. 2). For alder, the difference in means among topsoil and subsoil was the greatest (see Fig. 2) and was significant (*p*-value of the *t*-test = 0.007). For oak, significant difference in means between topsoil and subsoil was also measured (*p*-value = 0.018). For spruce, no significant difference between means among topsoil and subsoil was noted (*p*-value = 0.235).

3.2. Pore size fractions and bulk density of soils at the three forest stands

The mean content of macropores (>300 μm) for alder stand in topsoil was significantly higher than for spruce and also than for oak (see Table 1). No significant differences in total porosities among tested tree species were observed for topsoil. For subsoil, also no significant differences in content of macropores and total porosity were noted (Table 1). For topsoil in spruce stand, significantly higher contents of upper size capillary pores (in range 90–300 and also in range 30–90 μm) were noted. For each tree species, total porosity and macroporosity in topsoil were higher than those in subsoil (see Table 1).

For topsoil, the lowest mean bulk density was observed in alder stand (see Fig. 3). Only slightly higher mean value was measured in spruce stand. The highest value for bulk density was observed in oak stand. Differences among alder and oak stands were significant (Fig. 3). For topsoil, the broadest range and variance of measured values were observed in spruce stand. The situation was different for subsoil. The greatest bulk density was measured in alder soil and the lowest value in spruce soil (see significant differences in Fig. 3).

3.3. Dye tracer infiltration tests and preferential flow

Dye tracer infiltration tests show that the geometry of preferential flow in the soil profiles differed distinctly among the tested forest stands (see Fig. 4). The greatest lateral flow in topsoil was observed for alder profile in the drilosphere/humus layer (large uninterrupted dyed area, see Fig. 4a and detail in Fig. S5a). In oak profile, by contrast, water

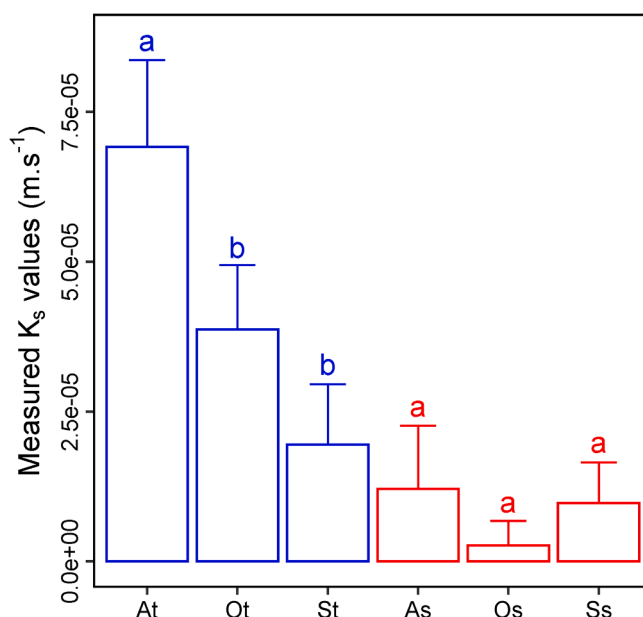


Fig. 2. Field saturated hydraulic conductivity (K_s) measured in the three different tree monocultures (mean and SD, $n = 4$) — alder (A), oak (O), spruce (S) — in topsoil (t, blue) and subsoil (s, red). Different letters over barplots indicate significant differences in means (for topsoil and subsoil separately) according to Tukey's HSD test.

Table 1

Content of pores (mean $\pm$ SD, $n = 10$) and corresponding ranges of equivalent pore diameters. Different letters following percentages indicate significant differences in means (for topsoil and subsoil separately) for each pore size range according to Tukey's HSD test.

Equivalent diameters of pores (μm)			>300	From 90 to 300	From 30 to 90	Total porosity*
Content of pores (% of total sample volume)	Topsoil	Alder	13.2 $\pm$ 6.2a	6.2 $\pm$ 0.8b	2.4 $\pm$ 0.4b	64.3 $\pm$ 3.0a
		Oak	4.2 $\pm$ 3.5b	7.2 $\pm$ 1.5b	2.6 $\pm$ 0.5b	61.5 $\pm$ 3.4a
		Spruce	7.5 $\pm$ 3.0b	11.7 $\pm$ 4.1a	5.3 $\pm$ 2.1a	63.4 $\pm$ 3.8a
	Subsoil	Alder	4.8 $\pm$ 2.8a	4.8 $\pm$ 1.2a	2.4 $\pm$ 0.4a	59.9 $\pm$ 3.3a
		Oak	3.0 $\pm$ 1.5a	3.4 $\pm$ 1.1b	1.9 $\pm$ 0.5b	57.3 $\pm$ 2.2a
		Spruce	5.5 $\pm$ 2.1a	4.5 $\pm$ 0.8ab	2.5 $\pm$ 0.4a	57.8 $\pm$ 1.9a

*Total porosity includes all pore diameters.

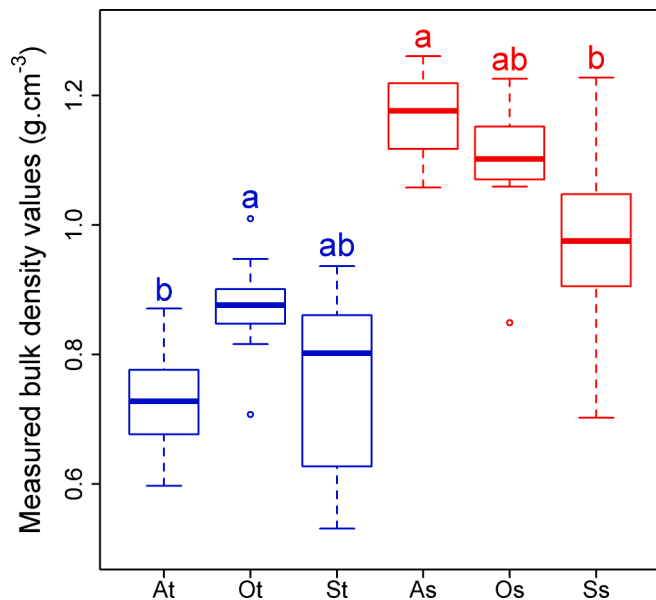


Fig. 3. Boxplot (sample minimum, first quartile, median, third quartile, maximum, $n = 10$) of bulk density measured in the three different tree monocultures — alder (A), oak (O), spruce (S) — in topsoil (t, blue) and subsoil (s, red). Different letters over boxplots indicate significant differences in means (for topsoil and subsoil separately) according to Tukey's HSD test. Empty circles are outliers (>1.5 times the interquartile range from the box).

flowed directly (with the least horizontal flow in topsoil, Table 2 and detail in Fig. S5b) to the deeper layers by distinct preferential pathways along strong roots (see dark blue area in Fig. 4b). In spruce profile, the smallest maximum vertical and lateral (horizontal) ranges for flow of the dyed water were observed (see Table 2 and Fig. 4c). The fermentation layer of spruce (Fig. S1) absorbed a large amount of dyed water (see detail in Fig. S3b), thereby slowing water flow to the deeper layers (see the longest infiltration time of the dyed water in spruce stand in Table 2). In alder and also in oak stand, preferential flow in earthworm burrows was observed (see Fig. S4). In spruce stand, preferential flow in earthworm burrows was not observed.

Similarly to the findings described above, semiquantitative evaluation of the dye tracer distribution patterns also differed markedly between stands (Table 3). For the alder stand, the lowest percentage of observation points without dye was observed (see macroscale in Table 3) and high percentage of dyed patches was located in claystone cracks. Further microscopic inspection of these stained cracks on thin soil sections, however, showed that all of them were connected (according to infiltrated dyed water pathways) to the drilosphere and/or roots, while the other ends of the cracks often were dead-ends. The fact that there also exist cracks that are not conductive for dyed water is supported by microscopic observation of clay microsities (studied by thin soil sections in microscale) outside roots or drilosphere that contain no stained patches despite presence of cracks. On the contrary, the cracks

were generally dyed in areas of soil surrounding roots or drilosphere (see high number of dyed cracks around root and burrow microsities of Table 3). Microscopic observation also revealed that dyed root channels are often connected to dyed worm-created structures (see 59 % and 34 % of drilosphere, respectively, for alder-root and oak-root microsities in Table 3). Therefore, at the oak stand, a similar pattern as for alder can be seen in microscale. In macroscale, differently from alder and spruce stands, dyed drilosphere in oak stand was more abundant than dyed cracks (see macroscale in Table 3). At the spruce stand, earthworm activity was not observed, and therefore the relative importance of dyed cracks was high. As described above, all the stained cracks can be traced to roots. The microscopic observation of thin soil sections revealed that cracks in the vicinity of roots contained many more stained patches than did cracks in clay microsities at the spruce stand.

3.4. Abundance of earthworms

For alder stand, the total number of earthworms was higher than for oak and for spruce stand, where the lowest abundance of individuals was noted (see Table S1). Distinct difference in means among alder and spruce stands was noted. Only one earthworm species (endogeic ecological group) was observed in spruce stand. Species from anecic and endogeic ecological groups but no species from epigeic group were observed in oak stand. The highest number of species and all three main ecological groups of earthworms (epigeic, endogeic, and anecic) were noted for alder stand (Table S1).

4. Discussion

In agreement with our hypothesis, we found that the soil under alder trees with the most readily decomposable litter had the highest values of K_s . This was accompanied by the greatest macroporosity (pores $> 300 \mu\text{m}$) and lowest bulk density values in the topsoil of alder stand. These values correspond with the highest infiltration rate that is caused by enhanced preferential flow in soils under alder trees. In agreement with our hypothesis, preferential flow in the alder stand was enhanced by intensive activity of earthworms, which prefer sites with litter rich in nitrogen and low in lignin and polyphenols (Ashwood et al., 2017; Hendriksen, 1990; Frouz et al., 2013; Rajapaksha et al., 2013). This corresponds with the lowest C/N ratio for alder litter (Frouz et al., 2013). The abundance of earthworms and number of species was highest in the alder stand, intermediate in oak, and lowest in soil under spruce. This trend is in agreement with results obtained by Walmsley et al. (2019) and Frouz et al. (2013), who have also sampled earthworms at this spoil heap and in the individual tree monocultures.

In agreement with previous studies, our results highlight the importance of earthworms for generating preferential flow at post-mining sites (Scullion and Malik, 2000; Frouz and Kuráz, 2014) and elsewhere (Edwards et al., 1993). The earthworm effect varies in topsoil and subsoil. Epigeic earthworms were observed only in the alder stand, where they contributed to rapid decomposition of litter and as a consequence have distinctly influenced infiltration capacity and preferential flow in the topsoil (Ernst et al., 2009). Endogeic earthworms

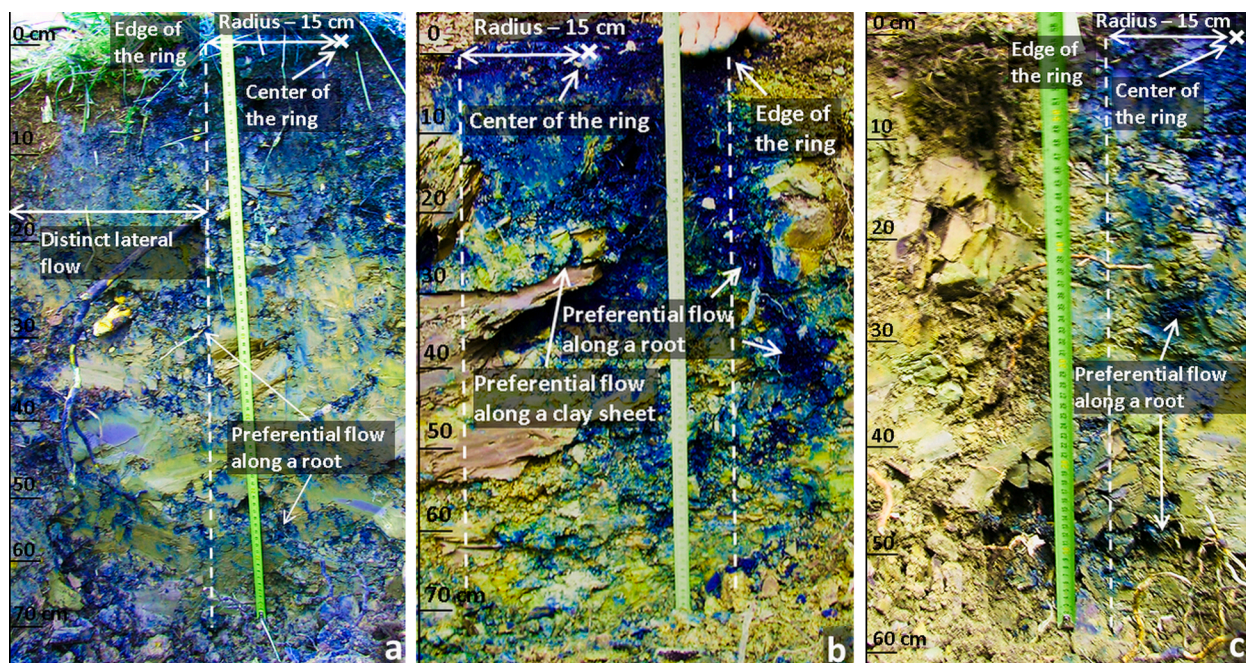


Fig. 4. Vertical cross section through the soil profile made in the center of the ring with infiltrated dye in the three different tree monocultures — (a) alder (left), (b) oak (center), and (c) spruce (right).

Table 2

Characteristics of dyed soil profiles for the tested tree species. Maximum vertical depth denotes spread of the dye from the soil surface. Maximum horizontal width denotes spread of the dye from the edge of the ring infiltrometer.

Tree species	Infiltration time for 24 L (min)	Max. vertical depth (cm)	Max. horizontal width (cm)	Maximum horizontal width (cm) at selected vertical depths (cm)		
				9 cm	13 cm	25 cm
Alder	5	90	30 (25)*	30	25	30
Oak	17	70	55 (32)*	5	5	40
Spruce	29	60	20 (25)*	15	15	20

*Corresponding vertical depths of cross sections are in brackets.

Table 3

Percentages (mean $\pm$ SD, $n = 3$) of individual structures measured as intercept of these structures with regular grid of sampling points. A 5×5 cm grid of sampling points was projected on soil profile photo in macroscale. In microscale, a 2×2 mm grid was projected on thin soil section prepared from samples of individual microhabitats.

	No dye	Drilosphere	Rhizosphere	Cracks
<i>Macroscale</i>				
Alder (A)	21.9 $\pm$ 5.6	16.0 $\pm$ 11.6	4.4 $\pm$ 1.1	56.9 $\pm$ 16.0
Oak (O)	51.0 $\pm$ 22.9	26.5 $\pm$ 7.9	2.3 $\pm$ 1.7	14.9 $\pm$ 18.4
Spruce (S)	36.3 $\pm$ 17.0	0.0 $\pm$ 0.0	33.8 $\pm$ 17.0	29.9 $\pm$ 5.6
<i>Microscale</i>				
A-clay	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
A-root	16.3 $\pm$ 4.9	58.6 $\pm$ 1.6	6.1 $\pm$ 0.5	18.7 $\pm$ 3.0
A-burrow	39.1 $\pm$ 17.8	48.7 $\pm$ 17.6	2.9 $\pm$ 1.5	8.3 $\pm$ 0.5
O-clay	100.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0	0.0 $\pm$ 0.0
O-root	48.6 $\pm$ 4.7	33.8 $\pm$ 6.0	1.5 $\pm$ 0.7	16.1 $\pm$ 0.6
O-burrow	25.3 $\pm$ 4.4	55.6 $\pm$ 9.6	0.6 $\pm$ 0.8	18.6 $\pm$ 8.7
S-clay	99.4 $\pm$ 0.9	0.4 $\pm$ 0.6	0.0 $\pm$ 0.0	0.2 $\pm$ 0.3
S-root	22.2 $\pm$ 5.1	0.4 $\pm$ 0.3	36.3 $\pm$ 5.7	41.0 $\pm$ 3.4

then incorporated fragments of litter into the mineral soil and formed a distinct layer of aggregates in the humus horizon of the soil (Frouz et al., 2007; Frouz, 2018, Fig. S1).

Infiltrated water then flows in macropores between these aggregates and through semi-permanent horizontal burrows of endogeic earthworms (Capowiez et al., 2014; van Schaik et al., 2014; Fig. S5a). In deeper soil layers, more permanent vertical burrows formed by anecic earthworm species, together with deeper tree roots, play a principal role for preferential flow of water to the deeper soil layers and subsoil, although anecic earthworms also contribute to aggregate formation in the topsoil (Tomlin et al., 1995; Pitkänen and Nuutinen, 1998; Capowiez et al., 2015; Fig. S4). This could be also seen on the thin soil sections, where larger macropores – anecic worm burrows – were distinguished by dark blue lining, presumably due to large volumes of dyed water flowing through and infiltration of water into the semi-permeable burrow walls. Earthworm burrows not only allow water percolation itself but they also connect otherwise blinded cracks in mudstones with percolating water and allow these cracks to participate in preferential water flow. Moreover, water can then be stored in these connected and opened cracks if the other ends of cracks are dead-ended. Part of this stored water can then slowly enter the clay soil matrix, where it is stored for a longer period. This effect may increase water retention during a dry period. Our results show that not only overall porosity but especially macropores and their connectivity can be crucial for infiltration and preferential water flow in real soils.

In agreement with our initial hypothesis, the slowest infiltration rate, lowest K_s values, and smallest range of the preferential flow were found in topsoil of spruce stand, which was characterized by the highest C/N ratio of the litter among the tested tree species (see Frouz et al., 2013) and the lowest total abundance and species number of earthworms. Hydrological effects observed for spruce stand can be caused by the following mechanisms: 1) Spruce needles have water-repellent properties (Bens et al., 2007; Rothe et al., 2002), which can slow down water entry into the soil. 2) The lack of nitrogen (high C/N ratio) and other specific properties of spruce litter (e.g., high content of essential oils) and, as a consequence, lack of earthworms, slow down litter decomposition and result in development of a fermentation layer (Prietzel and Bachmann, 2012; Frouz et al., 2013; Gurmessa et al., 2013; Frouz, 2018).

The typical spruce fermentation layer with partially decomposed needles and with large number of fine roots is bound with a network of fungi. These characteristics of fermentation layer presumably cause high tortuosity of the porous system. As a consequence, this layer only slowly transmits water to the deeper soil layers and water is accumulated here. This is also confirmed by dark blue color of the fermentation layer, caused by accumulation of dyed water (Fig. S3b). Under this layer there was a rapid transition to mineral soil formed by relatively impermeable clay, lacking soil aggregates and earthworms' burrows, and with few vertical roots penetrating the deeper soil layers (Fig. 4c).

In soil under oak, thickness of the drilosphere layer was only 3–5 cm (Fig. S1b) but there was no fermentation layer. In oak profile, therefore, water flows quickly (with the least lateral flow in topsoil, see Fig. S5b) to the deeper layers by preferential pathways along roots (Fig. 4b). Oak's strong root system, growing through clay sheets and earthworm burrows (see Fig. S4b), which often can be found together, presumably creates these pathways. Similar pathways are also present in mineral layer under the humus horizon in alder forest. These results are in agreement with findings from the literature that root architecture plays an important role in water distribution within the soil profile, especially to the deeper layers (Devitt and Smith, 2002; Luo et al., 2019; Zhu et al., 2019). While oak generally has a relatively long vertical tap root with lateral roots branching off (heart or tap root type), spruce roots, on the other hand, have a plate-like distribution (plate type), with the majority of roots at shallow depths (Danjon et al., 1999; Dupuy et al., 2007). Many alder tree roots also have been found at greater depths, indicating that its root system is more similar to that of oak than spruce. Kautz et al. (2014) found that in a perennial cropping system, one-third of anecic earthworms were utilizing existing biopores while another one-third had created new biopores. The overburden material, which forms soils at these sites, has extreme textural properties (Roják 2013), which makes it different to most agricultural soils. The presence of solid clay slabs also means that it does not present an environment suitable for anecic earthworm species in the initial phases, as there are no deep roots that would penetrate the soil and create channels that the worms could utilize (Ruiz et al., 2015). This theory is also supported by the fact that anecic earthworm species were not present at the study sites prior to year 2000, even though several epigeic and endogeic species had colonized these sites shortly after reclamation (Pižl, 2001; Walmsley et al., 2019). At other post-mining sites, anecic species have been found to colonize the sites shortly after heaping (Dunger 1989; Dunger et al. 2001), so it is not likely that their absence is caused by their lower migration abilities.

Inasmuch as the overburden is expected to be very similar at these three sites (due to their adjacent position), we believe that the hydrophysical characteristics of the soil have been modified by the tree species directly via root growth and indirectly as their tree litter has facilitated growth of various soil organisms. These soil modifications then work in favor of the tree and create biostructures that support either water accumulation in the top layers of soil (in the case of spruce) or water flow into deeper soil layers along strong roots and earthworm burrows (in the cases of oak and alder). Therefore, water is directed to soil areas with high density of fine roots where it is needed most. According to our findings, deciduous broad-leaved trees (oak and alder) exhibited greater preferential flow, infiltration capacity, and heterogeneity of water flow field than did spruce. Therefore, selection of trees for forest reclamation seems to be important for the development of the emerging ecosystem, as it can strongly affect soil–water distribution, which is a problematic aspect on the clay spoil heap (Kuráž, 2001; Frouz and Kuráž, 2014; Cejpek et al., 2018).

Planting of individual tree species has mostly been abandoned in reclamation practice since the time when the studied “common garden experiment” has been set up and planting of mixtures of different tree species is now a preferred option. However, due to the economic benefits of rapid timber production, coniferous species are still being planted in large numbers. Our current research, together with findings of Frouz

et al. (2009), Frouz et al. (2013) and Walmsley et al. (2019) shows, that higher ratio of deciduous species would benefit the soil development. Soils with a well-developed humus layer and high infiltration capacity would then support biomass production in trees and increase the economic value of tree plantations at the post-mining sites.

For an investigation of the different tree species effect on infiltration and preferential flow, we selected three small sites, which are adjacent. This selection allowed us to study effects of tree monocultures on practically the same substrate. Clay soils have very specific properties, e.g. shrinking and swelling phenomena depending on the temporal change in soil water content, which can greatly affect soil hydraulic properties. In order to minimize these temporal changes for correct comparison of the sites, field work was restricted to two consecutive days. Therefore, it has to be mentioned here that the described hydrological effects of trees may not be valid for other locations on spoil heaps characterized by different substrate and also for different antecedent conditions of soil water content. Dye tracer infiltration tests were performed using constant ponding depth of 4 cm. Although this ponding depth is very similar to other studies using dye tracer infiltration (e.g. Schwen et al., 2014; Jačka et al., 2018; Zhu et al., 2019) and was the lowest possible to flood the whole infiltration surface (due to the terrain irregularities on the unmodified soil surface), size of preferential flow could have been amplified by small positive pressure head on the soil surface. During most rainfall events, the soil surface is not flooded, topsoil is less saturated by water and preferential flow fields may be different.

5. Conclusions

Soil recovery is essential for ecosystem development in post-mining areas. The initial way of treating such an area (e.g., planting trees) substantially determines properties of the emerging soil and likelihood of a successful recovery (Walmsley et al., 2019). Infiltration parameters, preferential flow, and related soil properties in the examined spoil heap were distinctly affected by the planted tree species. The highest number of earthworms and quantity of macropores, thickest organic/humus layer, highest K_s values and extent of horizontal preferential flow in topsoil and greatest infiltration capacity were observed at the alder stand. At the oak stand, considerable preferential flow along strong roots and the greatest horizontal range were observed in mineral subsoil. The least infiltration capacity, size of preferential flow, and number of earthworms were observed in the spruce stand, where the water flow was partially blocked in the specific fermentation layer.

Macropores often served as preferential flow paths. Trees affected macropore content and connectivity directly by their specific root system (channels around roots) and indirectly by their specific litter amount and quality, which is an important factor for occurrence of drilosphere. Earthworms contributed to the porosity around the roots, opened blinded cracks in claystones and formed a layer of aggregates in topsoil. Connection of root system to cracks in clay and to drilosphere also played a very important role for infiltration and water distribution.

The presented findings suggest that selection of tree species to be planted in initial stages of mine rehabilitation may have great consequences in terms of soil hydrology and soil formation generally. However, further research should be directed to the future development of hydrological properties of this emerging soil under individual monocultures. Further attention should also be paid to different locations and antecedent soil moisture conditions. Ideally, this research would be expanded to other tree species and mixed stands that are desirable from the ecological point of view.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

This research was supported by the Ministry of Agriculture of the Czech Republic (project no. QK1710241, Optimization of management of forest restoration on sites affected by surface mining); and the Internal Grant Agency of the Faculty of Environmental Sciences, Czech University of Life Sciences Prague (project no. 20174223 (4220013123127)). We would like to thank Lenka Linzmayerová, Jan Kožušník, David Votava, Jakub Smrček, and Fabio Vincentini for their help during field sampling and Jitka Hubáčová (SoWa Biology Centre) for evaluation of thin soil sections and profile photographs. We would also like to express our gratitude to the anonymous reviewers for their helpful comments.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.geoderma.2021.115372>.

References

- Alaoui, A., Helbling, A., 2006. Evaluation of soil compaction using hydrodynamic water content variation: Comparison between compacted and non-compacted soil. *Geoderma* 134, 97–108. <https://doi.org/10.1016/j.geoderma.2005.08.016>.
- Ali, G., Macrae, M., Walker, M., Laing, J., Lobb, D., 2018. Preferential flow in vertisolic soils with and without organic amendments. *Agric. Environ. Lett.* 3, 1–5. <https://doi.org/10.2134/aer2018.04.0018>.
- Alaoui, A., Caduff, U., Gerke, H.H., Weingartner, R., 2011. Preferential flow effects on infiltration and runoff in grassland and forest soils. *Vadose Zone J.* 10 (1), 367–377. <https://doi.org/10.2136/vzj2010.0076>.
- Andriuzzi, W.S., Bolger, T., Schmidt, O., 2013. The drilosphere concept: fine-scale incorporation of surface residue-derived N and C around natural *Lumbricus terrestris* burrows. *Soil Biol. Biochem.* 64, 136–138. <https://doi.org/10.1016/j.soilbio.2013.04.016>.
- Ashwood, F., Butt, K.R., Doick, K.J., Vanguelova, E.L., 2017. Investigating tree foliar preference by the earthworms *Aporrectodea longa* and *Allolobophora chlorotica* in reclaimed and loam soil. *Appl. Soil Ecol.* 110, 109–117. <https://doi.org/10.1016/j.apsoil.2016.10.007>.
- Bargués Tobella, A., Reese, H., Almaw, A., Bayala, J., Malmer, A., Laudon, H., Ilstedt, U., 2014. The effect of trees on preferential flow and soil infiltrability in an agroforestry parkland in semiarid Burkina Faso. *Water Resour. Res.* 50 (4), 3342–3354. <https://doi.org/10.1002/2013WR015197>.
- Benegas, L., Ilstedt, U., Rounsard, O., Jones, J., Malmer, A., 2014. Effects of trees on infiltrability and preferential flow in two contrasting agroecosystems in Central America. *Agric. Ecosyst. Environ.* 183, 185–196. <https://doi.org/10.1016/j.agee.2013.10.027>.
- Bens, O., Wahl, N.A., Fischer, H., Hüttl, R.F., 2007. Water infiltration and hydraulic conductivity in sandy cambisols: impacts of forest transformation on soil hydrological properties. *Eur. J. For. Res.* 126, 101–109. <https://doi.org/10.1007/s10342-006-0133-7>.
- Beven, Keith J., Clarke, Robin T., 1986. On the variation of infiltration into a homogeneous soil matrix containing a population of macropores. *Water Resour. Res.* 22 (3), 383–388. <https://doi.org/10.1029/WR022i003p00383>.
- Beven, Keith, Germann, Peter, 1982. Macropores and water flow in soils. *Water Resour. Res.* 18 (5), 1311–1325. <https://doi.org/10.1029/WR018i005p01311>.
- Bonta, J.V., Amerman, C.R., Harlukowicz, T.J., Dick, W.A., 1997. Impact of coal surface mining on three Ohio watersheds - surface-water hydrology. *JAWRA* 33, 907–917. <https://doi.org/10.1111/j.1752-1688.1997.tb04114.x>.
- Borůvka, L., Kozák, J., Mühlhanslová, M., Donátová, H., Nikodem, A., Němeček, K., Drábek, O., 2012. Effect of covering with natural topsoil as a reclamation measure on brown-coal mining dumpsites. *J. Geochem. Explor.* 113, 118–123. <https://doi.org/10.1016/j.gexplo.2011.11.004>.
- Bouché, M.B., 1975. Action de la faune sur les états de lamatière organique dans les écosystèmes, in: Kilbertus, G., Reisinger, O., Mourey, A., Canela da Fonseca, J.A. (Eds.), *Humification et biodegradation*. Pierron, Sarreguemines, France, pp. 157–168.
- Bradshaw, A.D., 1983. The reconstruction of ecosystems. *J. Appl. Ecol.* 20, 1–27. <https://doi.org/10.2307/2403372>.
- Bradshaw, Anthony, 1997. Restoration of mined lands - using natural processes. *Ecol. Eng.* 8 (4), 255–269. [https://doi.org/10.1016/S0925-8574\(97\)00022-0](https://doi.org/10.1016/S0925-8574(97)00022-0).
- Brown, George G., Barois, Isabelle, Lavelle, Patrick, 2000. Regulation of soil organic matter dynamics and microbial activity in the drilosphere and the role of interactions with other edaphic functional domains. *Eur. J. Soil Biol.* 36 (3–4), 177–198. [https://doi.org/10.1016/S1164-5563\(00\)01062-1](https://doi.org/10.1016/S1164-5563(00)01062-1).
- Capowiez, Yvan, Sammartino, Stéphane, Michel, Eric, 2014. Burrow systems of endogeic earthworms: effects of earthworm abundance and consequences for soil water infiltration. *Pedobiologia* 57 (4–6), 303–309. <https://doi.org/10.1016/j.pedobi.2014.04.001>.
- Capowiez, Yvan, Bottinelli, Nicolas, Sammartino, Stéphane, Michel, Eric, Jouquet, Pascal, 2015. Morphological and functional characterisation of the burrow systems of six earthworm species (Lumbricidae). *Biol. Fertil. Soils* 51 (7), 869–877. <https://doi.org/10.1007/s00374-015-1036-x>.
- Cheng, Yanyan, Oden, Fred L., Zhu, Jianting, 2017. Earthworms and tree roots: A model study of the effect of preferential flow paths on runoff generation and groundwater recharge in steep, saprolitic, tropical lowland catchments. *Water Resour. Res.* 53 (7), 5400–5419. <https://doi.org/10.1002/2016WR020258>.
- Cejpek, J., Kuráz, V., Frouz, J., 2013. Hydrological Properties of Soils in Reclaimed and Unreclaimed Sites after Brown-Coal Mining. *Polish J. Environ. Studies* 22, 645–652.
- Cejpek, J., Kuráz, V., Vinduškova, O., Frouz, J., 2018. Water regime of reclaimed and unreclaimed post-mining sites. *Ecohydrology* 11, e1911. <https://doi.org/10.1002/eco.1911>.
- Danjon, F., Sinoquet, H., Godin, C., Colin, F., Drexhage, M., 1999. Characterisation of structural tree root architecture using 3D digitising and AMAPmod software. *Plant Soil* 211, 241–258. <https://doi.org/10.1023/A:1004680824612>.
- Devitt, Dale A., Smith, Stanley D., 2002. Root channel macropores enhance downward movement of water in a Mojave Desert ecosystem. *J. Arid Environ.* 50 (1), 99–108. <https://doi.org/10.1006/jare.2001.0853>.
- Dunger, W., 1989. The return of the soil fauna to coal mined areas in the German Democratic Republic. In: Majer, J.D. (Ed.), *Animals in primary succession*. Cambridge University Press, New York, The role of fauna in reclaimed lands, pp. 307–337.
- Dunger, Wolfram, Wanner, Manfred, Hauser, H., Hohberg, K., Schulz, H.-J., Schwalbe, T., Seifert, B., Vogel, J., Voigtländer, K., Zimdars, B., Zülka, K.P., 2001. Development of soil fauna at mine sites during 46 years after afforestation. *Pedobiologia* 45 (3), 243–271. <https://doi.org/10.1078/0031-4056-00083>.
- Dupuy, Lionel, Fourcaud, Thierry, Stokes, Alexia, 2007. A numerical investigation into the influence of soil type and root architecture on tree anchorage. In: Stokes, Alexia, Spanos, Ioannis, Norris, Joanne E., Cammeraat, Erik (Eds.), *Developments in Plant and Soil Sciences Eco-and Ground Bio-Engineering: The Use of Vegetation to Improve Slope Stability*. Springer Netherlands, Dordrecht, pp. 175–189.
- Edwards, W.M., Shipitalo, M.J., Owens, L.B., Dick, W.A., 1993. Factors affecting preferential flow of water and atrazine through earthworm burrows under continuous no-till corn. *J. Environ. Qual.* 22, 453–457. <https://doi.org/10.2134/jeq1993.00472425002200030008x>.
- Eijkkelkamp, 2019. 08.01 Sandbox for pF-determination. User manual M-0801E. https://www.eijkkelkamp.com/download.php?file=M0801E_Sandbox_for_pf_determination_d575.pdf, 1/2019, accessed: 12/11/2020.
- Elrick, D.E., Reynolds, W.D., Tan, K.A., 1989. Hydraulic conductivity measurements in the unsaturated zone using improved well analyses. *Ground Water Monit. Rev.* 9, 184–193. <https://doi.org/10.1111/j.1745-6592.1989.tb01162.x>.
- Ernst, Gregor, Felten, Daniel, Vohland, Michael, Emmerling, Christoph, 2009. Impact of ecologically different earthworm species on soil water characteristics. *Eur. J. Soil Biol.* 45 (3), 207–213. <https://doi.org/10.1016/j.ejsobi.2009.01.001>.
- Frouz, J., 2018. Effects of soil macro- and mesofauna on litter decomposition and soil organic matter stabilization. *Geoderma* 332, 161–172. <https://doi.org/10.1016/j.geoderma.2017.08.039>.
- Frouz, J., 2014. *Soil Biota and Ecosystem Development in Post Mining Sites*. CRC Press Taylor & Francis, Boca Raton, FL, USA.
- Frouz, J., Kuráz, V., 2014. Soil fauna and soil physical properties. In: Frouz, J. (Ed.), *Soil Biota and Ecosystem Development in Post Mining Sites*. CRC Press Taylor & Francis, Boca Raton, FL, USA, pp. 265–278.
- Frouz, Jan, Kepřl, Beate, Pižl, Václav, Tajovský, Karel, Starý, Josef, Lukešová, Alena, Nováková, Alena, Balík, Vladimír, Háněl, Ladislav, Materna, Jan, Düker, Christian, Chalupský, Josef, Rusek, Josef, Heinkele, Thomas, 2001. Soil biota and upper soil layer development in two contrasting post-mining chronosequences. *Ecol. Eng.* 17 (2–3), 275–284. [https://doi.org/10.1016/S0925-8574\(00\)00144-0](https://doi.org/10.1016/S0925-8574(00)00144-0).
- Frouz, Jan, Elhottová, Dana, Kuráz, Václav, Souřková, Monika, 2006. Effects of soil macrofauna on other soil biota and soil formation in reclaimed and unreclaimed post mining sites: results of a field microcosm experiment. *Appl. Soil Ecol.* 33 (3), 308–320. <https://doi.org/10.1016/j.apsoil.2005.11.001>.
- Frouz, J., Pižl, V., Tajovský, K., 2007. The effect of earthworms and other saprophagous macrofauna on soil microstructure in reclaimed and un-reclaimed post-mining sites in Central Europe. *Eur. J. Soil Biol.* 43, 184–189. <https://doi.org/10.1016/j.ejsobi.2007.08.033>.
- Frouz, Jan, Pižl, Václav, Cienciala, Emil, Kalčík, Jiří, 2009. Carbon storage in post-mining forest soil, the role of tree biomass and soil bioturbation. *Biogeochem.* 94 (2), 111–121. <https://doi.org/10.1007/s10533-009-9313-0>.
- Frouz, Jan, Kalčík, Jiří, Velichová, Václava, 2011. Factors causing spatial heterogeneity in soil properties, plant cover, and soil fauna in a non-reclaimed post-mining site. *Ecol. Eng.* 37 (11), 1910–1913. <https://doi.org/10.1016/j.ecoleng.2011.06.039>.
- Frouz, J., Livečková, M., Albrechtová, J., Chronáková, A., Cajthaml, T., Pižl, V., Háněl, L., Starý, J., Baldrian, P., Lhotáková, Z., Šimáková, H., Cepáková, Š., 2013. Is the effect of trees on soil properties mediated by soil fauna? A case study from post-mining sites. *Forest Ecol. Manag.* 309, 87–95. <https://doi.org/10.1016/j.foreco.2013.02.013>.
- Frouz, J., Mudrák, O., Reitschmiedová, E., Walmsley, A., Vachová, P., Šimáková, H., Albrechtová, J., Moradi, J., Kučera, J., 2018. Rough wave-like heaped overburden promotes establishment of woody vegetation while leveling promotes grasses during unassisted post mining site development. *J. Environ. Manag.* 205, 50–58. <https://doi.org/10.1016/j.jenvman.2017.09.065>.
- Gerke, H.H., 2006. Preferential flow description for structured soils. *J. Plant Nutr. Soil Sci.* 169, 382–400. <https://doi.org/10.1002/jpln.200521955>.
- Gerke, Horst H., Köhne, J. Maximilian, 2002. Estimating hydraulic properties of soil aggregate skins from sorptivity and water retention. *Soil Sci. Soc. Am. J.* 66 (1), 26–36. <https://doi.org/10.2136/sssaj2002.2600>.

- Gurmesa, G.A., Schmidt, I.K., Gundersen, P., Vesterdal, L., 2013. Soil carbon accumulation and nitrogen retention traits of four tree species grown in common gardens. *Forest Ecol. Manag.* 309, 47–57. <https://doi.org/10.1016/j.foreco.2013.02.015>.
- Hallam, Jamal, Hodson, Mark E., 2020. Impact of different earthworm ecotypes on water stable aggregates and soil water holding capacity. *Biol. Fertil. Soils* 56 (5), 607–617. <https://doi.org/10.1007/s00374-020-01432-5>.
- Helliwell, J.R., Sturrock, C.J., Mairhofer, S., Craigon, J., Ashton, R.W., Miller, A.J., Whalley, R., Mooney, S.J., 2017. The emergent rhizosphere: imaging the development of the porous architecture at the root-soil interface. *Scient. Report.* 7, 14875. <https://doi.org/10.1038/s41598-017-14904-w>.
- Hendriksen, N.B., 1990. Leaf litter selection by detritivore and geophagous earthworms. *Biol. Fertil. Soils* 10, 17–21. <https://doi.org/10.1007/BF00336119>.
- Hillel, D., 1998. *Environmental Soil Physics*. Academic Press, San Diego, USA, p. 771.
- Horton, R.E., 1933. The role of infiltration in the hydrologic cycle. *Trans. Am. Geophys. Union* 14, 446–460. <https://doi.org/10.1029/TR014i001p00446>.
- Jačka, L., Pavlásek, J., Jindrová, M., Bašta, P., Černý, M., Balvín, A., Pech, P., 2012. Steady infiltration rates estimated for a mountain forest catchment based on the distribution of plant species. *J. For. Sci.* 58 (No. 12), 536–544. <https://doi.org/10.17221/57/2012-JFS>.
- Jačka, L., Pavlásek, J., Kuráz, V., Pech, P., 2014. A comparison of three measuring methods for estimating the saturated hydraulic conductivity in the shallow subsurface layer of mountain podzols. *Geoderma* 219–220, 82–88. <https://doi.org/10.1016/j.geoderma.2013.12.027>.
- Jačka, L., Pavlásek, J., Pech, P., Kuráz, V., 2016. Assessment of evaluation methods using infiltration data measured in heterogeneous mountain soils. *Geoderma* 276, 74–83. <https://doi.org/10.1016/j.geoderma.2016.04.023>.
- Jačka, L., Pavlásek, J., Bašta, P., Kovář, M., Kuráz, V., 2018. The layering of a mountain podzol can strongly affect the distribution of infiltrated water in the soil profile. *J. Hydrol. Hydrom.* 66, 348–356. <https://doi.org/10.2478/johh-2018-0014>.
- Jarvis, N.J., 2007. A review of non-equilibrium water flow and solute transport in soil macropores: Principles, controlling factors, and consequences for water quality. *Eur. J. Soil Sci.* 58, 523–546. <https://doi.org/10.1111/j.1365-2389.2007.00915.x>.
- Kautz, Timo, Lüsebrink, Marcel, Pätzold, Stefan, Vetterlein, Doris, Pude, Ralf, Athmann, Miriam, Küpper, Paul M., Perkons, Ute, Köpke, Ulrich, 2014. Contribution of anecic earthworms to biopore formation during cultivation of perennial leafy crops. *Pedobiologia* 57 (1), 47–52. <https://doi.org/10.1016/j.pedobi.2013.09.008>.
- Kříbek, Bohdan, Strnad, Mojmir, Boháček, Zbyněk, Šýkorová, Ivana, Čejka, Jiří, Sobalik, Zdeněk, 1998. Geochemistry of Miocene lacustrine sediments from the Sokolov Coal Basin (Czech Republic). *Int. J. Coal Geol.* 37 (3–4), 207–233. [https://doi.org/10.1016/S0166-5162\(98\)00002-0](https://doi.org/10.1016/S0166-5162(98)00002-0).
- Kuráz, V., 2001. Soil properties and water regime of reclaimed surface dumps in the North Bohemian brown-coal region — A field study. *Waste Manage.* 21, 147–151. [https://doi.org/10.1016/S0956-053X\(00\)00064-7](https://doi.org/10.1016/S0956-053X(00)00064-7).
- Kuráz, V., Frouz, J., Kuráz, M., Mako, A., Shustr, V., Cejpek, J., Romanov, O.V., Abakumov, E.V., 2012. Changes in some physical properties of soils in the chronosequence of self-overgrown dumps of the Sokolov quarry-dump complex, Czechia. *Eurasian Soil Sci.* 45, 266–272. <https://doi.org/10.1134/S1064229312030076>.
- Kutilek, M., Nielsen, D.R., 1994. *Soil Hydrology*. Catena Verlag, Cremlingen-Destedt, Germany, p. 370.
- Lim, T., Spokas, K., Feyereisen, G., Novak, J., 2016. Predicting the impact of biochar additions on soil hydraulic properties. *Chemosphere* 142, 136–144. <https://doi.org/10.1016/j.chemosphere.2015.06.069>.
- Luo, Ziteng, Niu, Jianzhi, Zhang, Linus, Chen, Xiongwen, Zhang, Wei, Xie, Baoyuan, Du, Jie, Zhu, Zhijun, Wu, Shanshan, Li, Xiang, 2019. Roots-enhanced preferential flows in deciduous and coniferous forest soils revealed by dual-tracer experiments. *J. Environ. Qual.* 48 (1), 136–146. <https://doi.org/10.2134/jeq2018.03.0091>.
- Marashi, A.R.A., Scullion, J., 2004. Porosity and hydrological changes in surface mine soils, paper 405. In: Raine, S.R., Biggs, A.J.W., Menzies, N.W., Freebairn, D.M., Tolmie, P.E. (Eds.), *ISCO 2004—13th International Soil Conservation Organisation Conference, Conserving Soil and Water for Society: Sharing Solutions*. Brisbane, Australia.
- Macdonald, S. Ellen, Landhäusser, Simon M., Skousen, Jeff, Franklin, Jennifer, Frouz, Jan, Hall, Sarah, Jacobs, Douglass F., Quideau, Sylvie, 2015. Forest restoration following surface mining disturbance: challenges and solutions. *New Forest.* 46 (5–6), 703–732. <https://doi.org/10.1007/s11056-015-9506-4>.
- Nobles, M.M., Wilding, L.P., Lin, H.S., 2010. Flow pathways of bromide and Brilliant Blue FCF tracers in caliche soils. *J. Hydrol.* 393 (1–2), 114–122. <https://doi.org/10.1016/j.jhydrol.2010.03.014>.
- Philip, J.R., 1957. The theory of infiltration: 1 the infiltration equation and its solution. *Soil Sci.* 83, 345–357. <https://doi.org/10.1097/00010694-200606001-00009>.
- Pitkänen, Jyrki, Nuutinen, Visa, 1998. Earthworm contribution to infiltration and surface runoff after 15 years of different soil management. *Appl. Soil Ecol.* 9 (1–3), 411–415. [https://doi.org/10.1016/S0929-1393\(98\)00098-5](https://doi.org/10.1016/S0929-1393(98)00098-5).
- Pízl, Václav, 2001. Earthworm Succession in Afforested Colliery Spoil Heaps in the Sokolov Region, Czech Republic. *Restor. Ecol.* 9 (4), 359–364. <https://doi.org/10.1046/j.1526-100X.2001.94004.x>.
- Prietz, J., Bachmann, S., 2012. Changes in soil organic C and N stocks after forest transformation from Norway spruce and Scots pine into Douglas fir, Douglas fir/spruce, or European beech stands at different sites in Southern Germany. *Forest Ecol. Manag.* 269, 134–148. <https://doi.org/10.1016/j.foreco.2011.12.034>.
- Rajapaksha, N.S.S., Butt, K.R., Vanguelova, E.L., Moffat, A.J., 2013. Earthworm selection of Short Rotation Forestry leaf litter assessed through preference testing and direct observation. *Soil Biol. Biochem.* 67, 12–19. <https://doi.org/10.1016/j.soilbio.2013.08.006>.
- Reynolds, W.D., Bowman, B.T., Brunke, R.R., Drury, C.F., Tan, C.S., 2000. Comparison of tension infiltrometer, pressure infiltrometer, and soil core estimates of saturated hydraulic conductivity. *Soil Sci. Soc. Am. J.* 64, 478–484. 10.2136/sssaj2000.642478x.
- Reynolds, W.D., 2008. Saturated hydraulic properties: Well permeameter. In: Carter, M. R., Gregorich, E.G. (Eds.), *Soil Sampling and Methods of Analysis*, 2nd ed. CRC Press Taylor & Francis, Boca Raton, pp. 1025–1042.
- Rojík, P., 2013. Geological Substrates and Heaping Process of Coal Mining Operations in the Sokolov Basin, Czech Republic. In: Frouz, J. (Ed.), *Soil Biota and Ecosystem Development in Post Mining Sites*. CRC Press Taylor & Francis, Boca Raton, pp. 1–18.
- Rothe, A., Kreutzer, K., Küchenhoff, H., 2002. Influence of tree species composition on soil and soil solution properties in two mixed spruce-beech stands with contrasting history in Southern Germany. *Plant Soil* 240, 47–56. <https://doi.org/10.1023/A:1015822620431>.
- Ruiz, Siul, Or, Dani, Schymanski, Stanislaus J., Balestrini, Raffaella, 2015. Soil penetration by earthworms and plant roots—mechanical energetics of bioturbation of compacted soils. *PLoS ONE* 10 (6), e0128914. <https://doi.org/10.1371/journal.pone.0128914>.
- Schwen, A., Backus, J., Yang, Y., Wendroth, O., 2014. Characterizing land use impact on multi-tracer displacement and soil structure. *J. Hydrol.* 519, 1752–1768. <https://doi.org/10.1016/j.jhydrol.2014.09.028>.
- Scullion, J., Malik, A., 2000. Earthworm activity affecting organic matter, aggregation and microbial activity in soils restored after opencast mining for coal. *Soil Biol. Biochem.* 32 (1), 119–126. [https://doi.org/10.1016/S0038-0717\(99\)00142-X](https://doi.org/10.1016/S0038-0717(99)00142-X).
- Shapiro, S.S., Wilk, M.B., 1965. An analysis of variance test for normality (complete samples). *Biometrika* 3, 591–611. <https://doi.org/10.2307/2333709>.
- Soilmoisture, 2012. Operating instructions. 2800 Guelph permeameter. https://www.soilmoisture.com/pdfs/Resource_Instructions_0898-2800_2800K1%20Guelph%20Permeameter%20.pdf, 12/2012, accessed: 12/11/2020.
- Soracco, C.G., Villarreal, R., Melani, E.M., Oderiz, J.A., Salazar, M.P., Otero, M.F., Irizar, A.B., Lozano, L.A., 2019. Hydraulic conductivity and pore connectivity. Effects of conventional and no-till systems determined using a simple laboratory device. *Geoderma* 337, 1236–1244. <https://doi.org/10.1016/j.geoderma.2018.10.045>.
- Tang, Qicheng, Duncan, Jonathan M., Guo, Li, Lin, Henry, Xiao, Dacheng, Eissenstat, David M., 2020. On the controls of preferential flow in soils of different hillslope position and lithological origin. *Hydrol. Process.* 34 (22), 4295–4306. <https://doi.org/10.1002/hyp.13883>.
- Tomlin, A.D., Shiptalo, M.J., Edwards, W.M., Protz, R., 1995. Earthworms and their influence on soil structure and infiltration. In: Press, C.R.C. (Ed.), *Earthworm ecology and biogeography in North America*. Boca Raton FL, pp. 159–184.
- van Schaik, Loes, Palm, Julian, Klaus, Zehe, Erwin, Schröder, Boris, 2014. Linking spatial earthworm distribution to macropore numbers and hydrological effectiveness. *Ecology* 7 (2), 401–408. <https://doi.org/10.1002/eco.1358>.
- Váška, J., Kuráz, V., Čermák, P., 1993. Numerical simulation of the surface runoff on the waste dump Radovesice (North Bohemia). *Proceedings of the Symposium “Advances in Water Science”*, Vol II, Stara Lesna, UHH SAV Bratislava, Slovakia, pp. 154–159.
- Wang, Kang, Zhang, Renduo, 2011. Heterogeneous soil water flow and macropores described with combined tracers of dye and iodine. *J. Hydrol.* 397 (1–2), 105–117. <https://doi.org/10.1016/j.jhydrol.2010.11.037>.
- Weiler, Markus, Naef, Felix, 2003. An experimental tracer study of the role of macropores in infiltration in grassland soils. *Hydrol. Process.* 17 (2), 477–493. <https://doi.org/10.1002/hyp.1136>.
- Weiler, Markus, Flüher, Hannes, 2004. Inferring flow types from dye patterns in macroporous soils. *Geoderma* 120 (1–2), 137–153. <https://doi.org/10.1016/j.geoderma.2003.08.014>.
- Walmsley, Alena, Vachová, Pavla, Hlava, Jakub, 2019. Tree species identity governs the soil macrofauna community composition and soil development at reclaimed post-mining sites on calcium-rich clays. *Eur. J. For. Res.* 138 (4), 753–761. <https://doi.org/10.1007/s10342-019-01202-5>.
- Zhang, San, Tang, G.H., Wang, WenQing, Li, Zen, Wang, Bo, 2020. Prediction and evolution of the hydraulic tortuosity for unsaturated flow in actual porous media. *Micropor. Mesop. Mat.* 298, 110097. <https://doi.org/10.1016/j.micromeso.2020.110097>.
- Zhang, Zhuanfang F., Groenevelt, Pieter H., Parkin, Gary W., 1998. The well shape factor for the measurement of soil hydraulic properties using the Guelph permeameter. *Soil Till. Res.* 49 (3), 219–221. [https://doi.org/10.1016/S0167-1987\(98\)00174-3](https://doi.org/10.1016/S0167-1987(98)00174-3).
- Zhu, X.A., Chen, C.F., Wu, J.N., Yang, J.B., Zhang, W.J., Zou, X., Liu, W.J., Jiang, X.J., 2019. Can intercrops improve soil water infiltrability and preferential flow in rubber-based agroforestry system? *Soil Tillage Res.* 191, 327–339. <https://doi.org/10.1016/j.still.2019.04.017>.